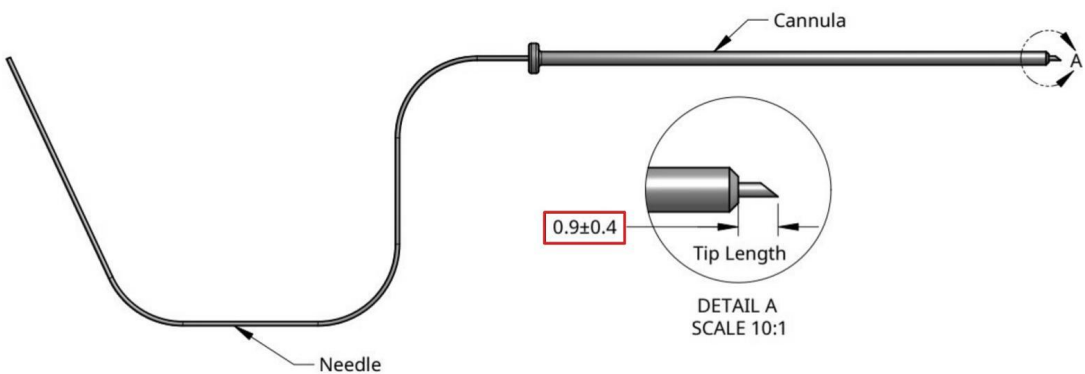


BROKEN QUALITY PRESS PRESENTS

The Needle & the Cannula

A case study in SPC network analysis

An essay by Jim Lehner



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The Needle & the Cannula

A case study in SPC network analysis

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First edition

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The Needle & the Cannula

Context

Since the release of its automated manufacturing lines to production, a manufacturer of insulin delivery technologies has seen an alarming volume of scrap. The scrap has become a particular concern at the Cannula- Needle Subassembly Stations, internally referred to as the CNSS (pronounced “cans”). The output of the CNSS is important because of their role as throughput limiters of the manufacturing process. Without the needle-cannula subassemblies, final product assembly cannot proceed.

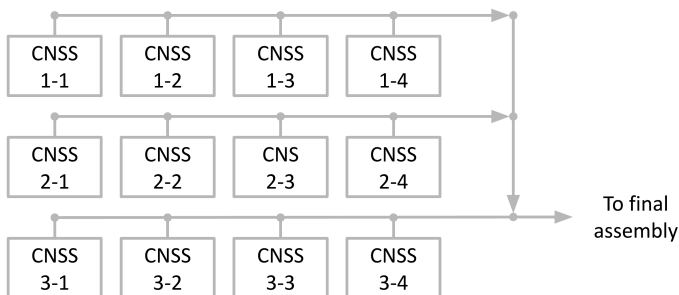


Figure 1. Schematic of manufacturing process composed of three lines, each line containing four Cannula-Needle Subassembly Stations (CNSS).

While the manufacturing team has made valiant efforts to reduce scrap, to

date, nothing seems to have worked. Adjustments that initially appeared to improve yields were quickly proven to be spurious. For every day the volume of scrap is low there are two days where it is high. For every day that production meets its targets, there are several when it's woefully deficient.

Given the state of this manufacturing process, what can the team do to facilitate process improvement? How can they take a process that has shown itself to be reliably deficient and turn it into a process that produces products of world-class quality? The answers to these questions lie in critical thinking that is guided by a novel approach to understanding variation called Statistical Process Control (SPC) network analysis.

Ready, aim, fix it

Regardless of a team's credentials and experience, if team members are not working to solve the same problem the chances of success are exceedingly low. Moreover, even if they are working to solve the same problem but that problem doesn't have a dominant financial impact on the business, improvement efforts will be in vain. Thus, the place to start any improvement effort is to identify the problem at hand. It is to discriminate between problems that need attention today and problems that can wait until tomorrow.

Rather than guessing where to begin, improvement efforts are most productive when guided by analysis. Given that "*the best analysis is the simplest analysis which gives the needed insight*,"¹ an ordered count of failures for the CNSS manufacturing process, like those shown in Figure 2, is illuminating.

¹ Donald J. Wheeler, *Understanding Variation*, (SPC Press, 1993), 61.

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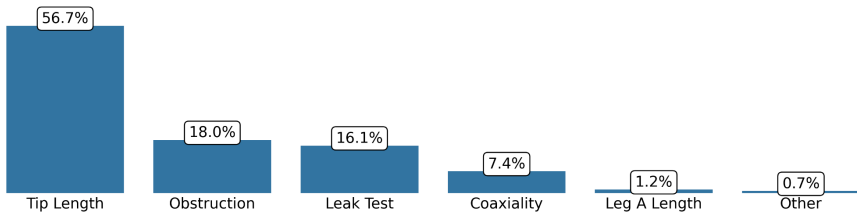


Figure 2. Bar chart of percent of failures by failure category.

By grouping and counting the number of failures in each failure category for the last month, the bar chart in Figure 2 reveals that 57% of cannula-needle subassembly failures were related to *tip length*. This amounts to a staggering 12,300 subassemblies that were scrapped. Extrapolated over the course of the year, 147,600 subassemblies will be discarded due to tip length failures. The cost of this scrap is staggering. Each subassembly costs \$3.35 (\$3.00 for the cannula and \$0.35 for the needle). The annual scrap-related financial loss is \$494,460 per year ($\$3.35 \times 147,600 = \$494,460$). This is a significant financial burden that warrants prompt investigation and corrective action by the manufacturing team.

An overview of tip length

Before collecting data and performing analysis it is important to remember that all data requires context. Without context data has no structure and tells no story. While 57% of cannula-needle subassemblies have been scrapped due to tip length failures in the last month—and, when extrapolated over the course of the year, this amounts to impressive losses—this information tells us nothing about the functional role that tip length plays in insulin delivery.

As shown in Figure 3, tip length is the measured distance between the shoulder of the soft cannula and the tip of the formed needle. After the needle is inserted into the cannula, the CNSS measures the distance from the shoulder of the cannula to the tip of the needle. The target tip length is 0.9 units with a tolerance of ± 0.4 units. This makes the Upper Specification Limit (USL) 1.3 units and the Lower Specification Limit (LSL) 0.5 units.

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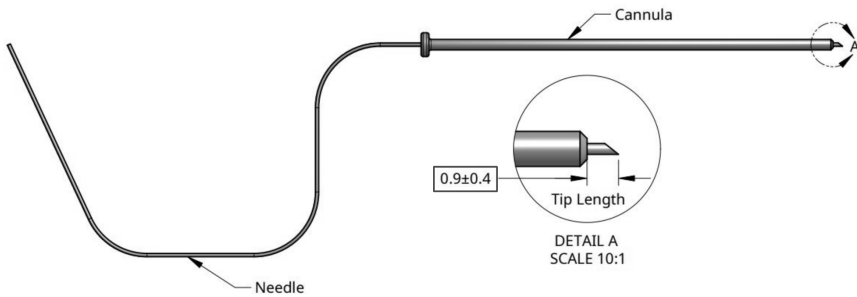


Figure 3. The target tip length for the cannula-needle subassembly is 0.9 units ± 0.4 units.

Tip length is a critical to quality (CTQ) dimension. It determines how deep the cannula is inserted into the subcutaneous tissue. If tip length is too long, the cannula may penetrate beyond the subcutaneous fat layer and into the underlying muscle tissue. Among other problems, this can lead to erratic insulin absorption rates and reduced insulin efficacy. If tip length is too short, the cannula may not reach the optimal depth. This can also lead to erratic insulin absorption or insulin leakage. Whether too long or too short, assemblies with tip lengths outside of specification are discarded because they cannot be reworked and they cannot reliably perform their designed function.

Network analysis of the current state

Given the prior context, it may be compelling to generate a process behavior chart using tip length data from one of the twelve CNSS. While there is nothing wrong with this approach, it is limited. The scope and scale of this manufacturing process demands a wider, more holistic view.

The typical application of process behavior charts is the sequential analysis of singular processes. Applied in the context of the CNSS manufacturing process, unique process behavior charts for each CNSS would be generated and their requisite insights acted upon. Although fruitful in its own right, this results in critical thinking and visual reasoning that is segmented. For example, the process behavior chart for CNSS-1-1 is evaluated and actions

are taken based on its insights. The process behavior chart for CNSS-1-2 is evaluated and actions are taken based on its insights. This process continues until all the CNSS have been evaluated and the requisite actions taken.

This partitioned analysis promotes a partitioned perspective. While collectively the CNSS operate as *a network of interdependent components working together to try to accomplish the aim of the system*, a sequential analysis of the CNSS evaluates each of its elements independently from the others. Moreover, the sequence of analysis, whether we start with CNSS-1-1 or start with CNSS-3-4, is arbitrary. As a rule, *we should never leave to guesswork what can be achieved through analysis*. Thus, rather than rely on the random and sequential analysis of each CNSS, a method that promotes a holistic view should be used. This method is called network analysis. The network analysis of the CNSS manufacturing process is shown in Figure 4.

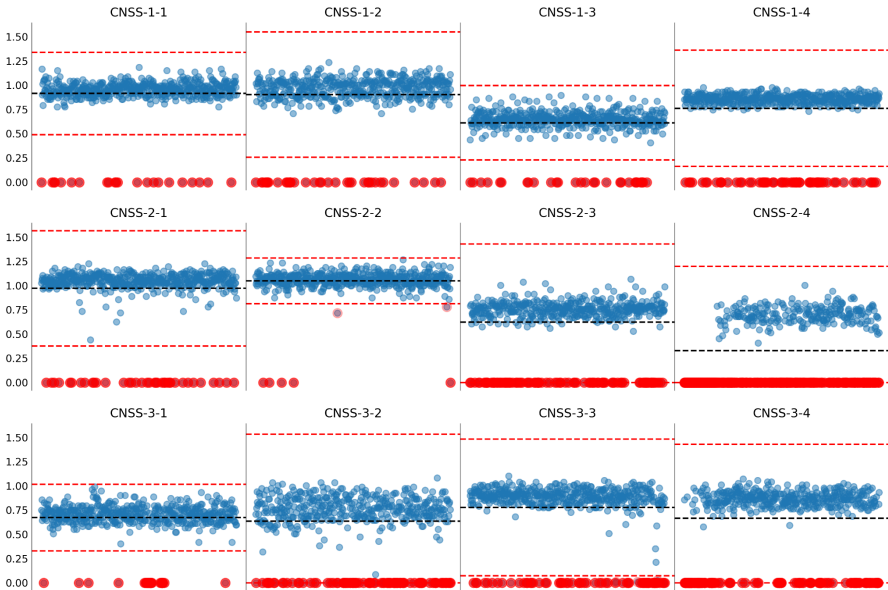


Figure 4. Network analysis of tip length data for the CNSS manufacturing process.

Network analysis is a grid-based data visualization technique that uses data-bountiful small-multiples. It facilitates uninterrupted visual reasoning across a network of interdependent components working together to accomplish an aim.

Each element in the grid of elements in a network analysis is the X chart portion of an XmR chart. The process limits for each grid element are calculated independently from the others. The shared y-axis of the grid elements enables direct visual comparison between the data in each grid element and the respective means (shown as black dotted lines) and process limits (shown as red dotted lines).

While it is clear that the means and process limits for each CNSS in Figure 4 are unique, the more compelling story is the characterizations. With a plenitude of values falling outside the process limits (highlighted in red) for every element in the grid, process behavior for all of the CNSS is characterized as unpredictable. This means that, regardless of which CNSS we choose to work on, improvement efforts will focus on eliminating the influence of assignable causes of exceptional variation.

Unlike common causes of routine variation, which are many in number but minimal in their effect, assignable causes of exceptional variation are few in number but dominate in their effect. When assignable causes are present, technical resources must work to eliminate their influence. Until this is achieved, efforts to center a process at its target or reduce common causes of routine variation will be in vain.

Although the influence of assignable causes appears to vary from line-to-line and machine-to-machine, a notable feature of the network analysis in Figure 4 is the consistent presence of assignable causes at zero. Displayed as a solid red line at the bottom of each X chart in Figure 4, whatever is causing the zero values appears to be influencing all 12 CNSS. If this is true, a CNSS agnostic solution to reducing zeros, something that could be applied to all 12 machines, may lie in wait.

The scale of the zero problem is given further context by the histogram in Figure 5. Here, the distribution of the tip length data for all twelve CNSS is presented in the context of the specification limits. Aside from the fact that

the distribution is not centered and, even if it was, its spread is wider than the tolerance, the left side of the figure is dominated by the stack of zeros.

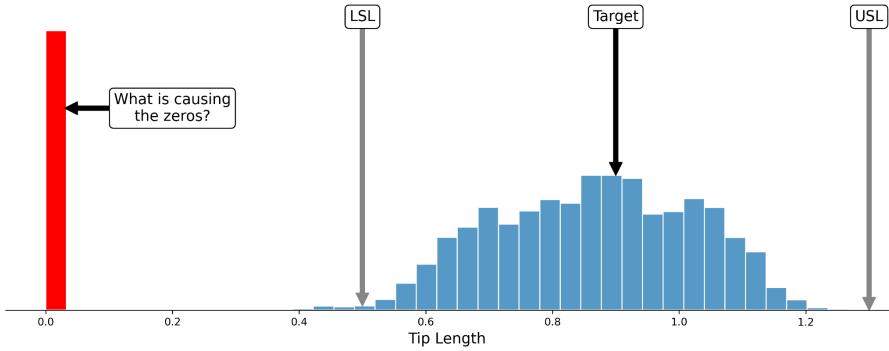


Figure 5. Capability histogram of CNSS tip length measurements.

A concerning but interesting insight, the stack of zeros accounts for 13.8% of tip length measurements for the entire manufacturing process. In service of identifying the underlying cause, the next step in this improvement effort is to understand the zeros by reviewing the tip length images associated with them.

Reviewing tip length images

Given the speed at which parts are produced by the CNSS, a visit to the manufacturing floor for direct observation of the assembly process reveals no insights. However, each tip length measurement is calculated from an image that can be reviewed on the organization's servers. Review of these images reveals that there are four conditions that produce a tip length measurement of zero. As shown in Figure 6, these conditions are stretched cannula, pierced cannula, excess lubricant, and nothing in the field of view (FOV). Of these conditions excess lubricant is the failure mode that reliably manifests across all CNSS, accounting for 83% of zeros.

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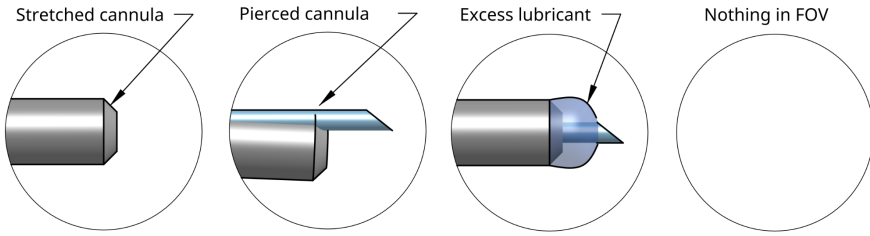


Figure 6: Tip length failure modes (from left to right): stretched cannula, pierced cannula, excess lubricant, and nothing in the field of view.

As is often the case with sliding surfaces, the lubricant used during the cannula–needle subassembly process reduces the friction between the two components. This allows the needle to slide into place more easily and without damaging the cannula. Without lubricant, the potential for stretching or piecing the cannula during the assembly process increases. Although one might assume that there is no such thing as too much lubricant, the excess lubricant image in Figure 6 is proof of otherwise. While it does not occur every time, the sequential buildup and bubbling of lubricant at the cannula–needle tip impedes the vision system from identifying the part features and registering the fiducials that are necessary for calculating tip lengths.

Searching for a solution

After discussing the lubrication problem internally, the manufacturing team proposes a solution. If the cannula and needle tip are obscured by excess lubricant and, as a result, tip length measurements are zero, why not remove it by some mechanical means? Better yet, why not avoid touching the assembly altogether and remove the excess lubricant using a burst of air? While testing is required to determine the parameters of the airburst and its efficacy, this solution can be evaluated on a single CNSS and, depending on the results, expanded to other machines in the manufacturing process.

After creating a test fixture and installing it on CNSS-1-1, a test of the

airburst system is performed on 250 subassemblies. The XmR chart² from the test is shown in Figure 7. Where prior to the addition of the airburst fixture CNSS-1-1 produced zeros 10% of the time, after the introduction of the airburst fixture CNSS-1-1 produced zeros 2.8% of the time. More importantly, a review of the images associated with the zeros indicates that excess lubricant is no longer a zero-generating failure mode in the sample of 250 measurements. Of the seven zero measurements, three show there was nothing in the field of view, three were the result of a stretched cannula, and one was the result of a pierced cannula. While these results do not guarantee the excess lubricant problem has been solved, it suggests that a viable mechanism for reducing the scrap caused by tip length has been discovered. That is, an assignable cause of exceptional variation has been eliminated.

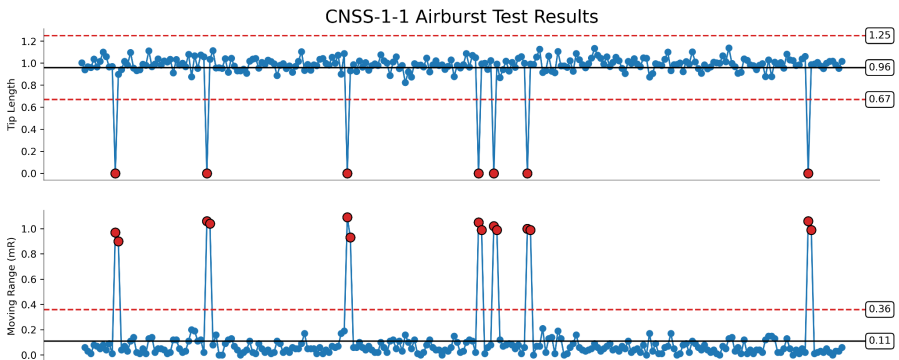


Figure 7: XmR chart of tip lengths after introducing the airburst test fixture to CNSS-1-1.

² An XmR chart plots the individual values on the X chart (top) and associated moving ranges on the mR chart (bottom).

Altering the process

Given the results of the airburst test, the design team gets to work finalizing the solution. While this effort, combined with the integration of the airburst system into production, takes a few weeks, the results shown in Figure 8 are telling.

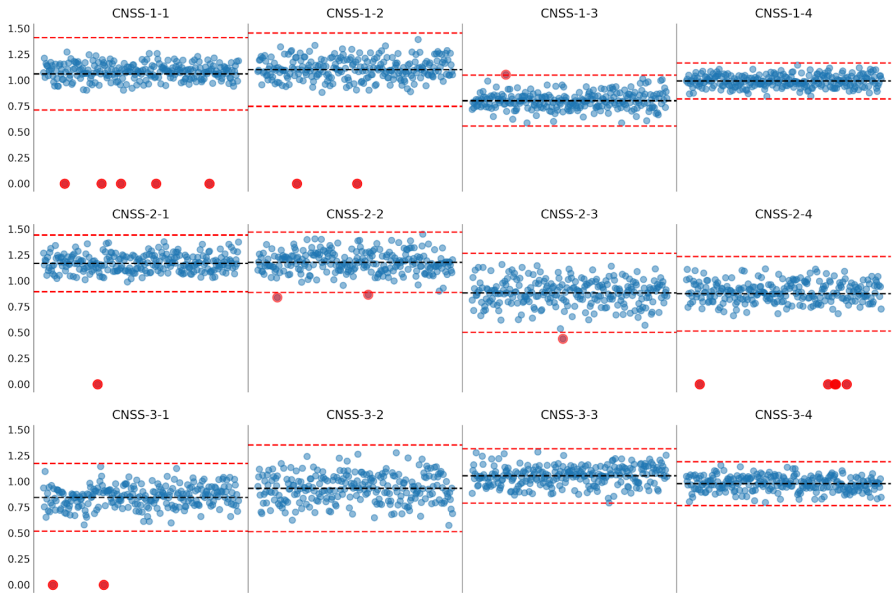


Figure 8: Network analysis of tip lengths after installing the airburst systems on all CNSS.

Unlike the baseline network analysis from Figure 4, where each element of the grid was visually dominated by a red line of zeros, only a handful of zero measurements remain after airburst integration. Moreover, the presence of zeros is no longer a uniform feature of all the CNSS. Some CNSS, like CNSS-1-3, CNSS-1-4, CNSS-3-2, CNSS-3-3, and CNSS-3-4, generated no zeros during this initial run. This suggests that a comprehensive improvement has been realized.

Adding to the reduction in zeros is the fact that the behavior of several of the CNSS is now characterized as predictable. A predictable process, as opposed to an unpredictable process, is influenced by only common causes of routine variation. Such characterization indicates that, moving forward, the team can expect tip lengths as long as the Upper Process Limit (UPL) or as short as the Lower Process Limit (LPL). That is, future process behavior can be predicted within the associated process limits. This sets the stage for additional improvement efforts. To understand what these follow-up efforts are and the order that they should be executed, understanding the process in terms of the voice of the customer is necessary. This is made possible with limit chart network analysis.

Limit chart network analysis

In contrast to the SPC based network analysis in Figures 4 and 8, where the grids are composed of X charts, limit chart network analysis creates a grid of run charts that places process data in the context of the specification limits and target.

In Figure 9, the target tip length is shown as a green dotted line, the specification limits are shown as gray dotted lines, and the means for each CNSS are shown as a black dotted lines. Where Figure 8 displays the tip lengths in the context of the voice of the process (the process limits), Figure 9 displays the tip lengths in the context of the voice of the customer (the specification limits). This context reveals several insights that Figure 8 obscured.

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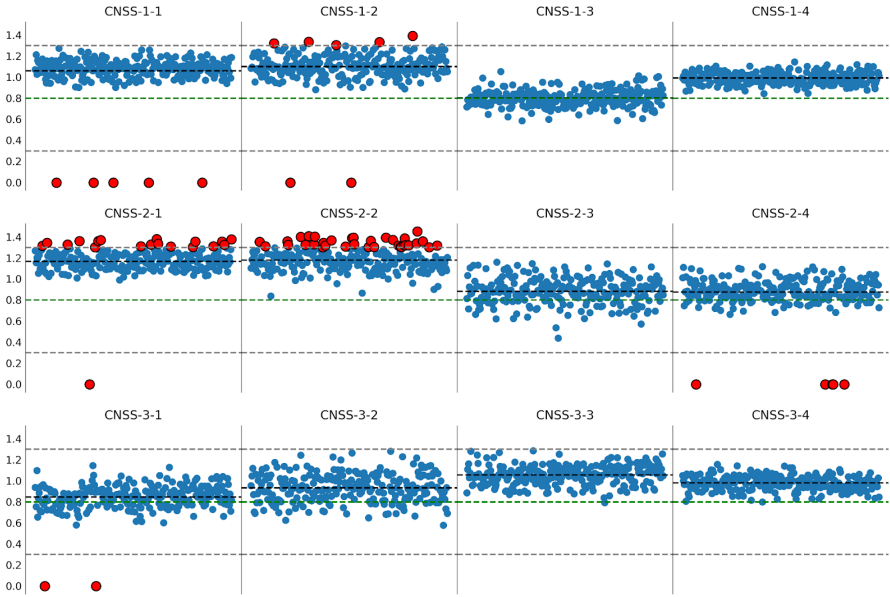


Figure 9: Limit chart network analysis of CNSS tip lengths after airburst installation.

Of note in Figure 9 is the divergence of the means from the target for several CNSS. For example, even though their behavior was characterized as predictable in Figure 8, CNSS-1-4, CNSS-3-2, CNSS-3-3, and CNSS-3-4 are all operating near the Upper Specification Limit (USL). To mitigate the production of non-conforming materials—i.e. tip lengths that are too long—all of the CNSS in the prior list would benefit from centering. That is, they would benefit from mechanical adjustments that align the mean of the distributions with the target.

Rather than make these adjustments haphazardly, the adjustment distance for each of the CNSS can be approximated by subtracting the target value from the mean. This yields a value called the centering distance.

$$\text{Centering distance} = \bar{X} - \text{Target}$$

The centering distances for the CNSS are shown in Figure 10. Sorted according to characterization, blue for predictable and red for unpredictable, and largest centering distance to smallest, the bar chart in Figure 10 imparts a logical sequence to the centering effort. It highlights that the four systems that are characterized as predictable can be centered immediately in accordance with their respective centering distances.

Given that all of the centering distances are positive, Figure 10 also reveals that the mean for each CNSS is “above” the target. Had a centering distance been negative, such a value would indicate the mean was below the target. While it is unlikely that this convention will map to mechanical adjustments in terms of up and down, it ensures that the adjustment logic is the same for each CNSS.

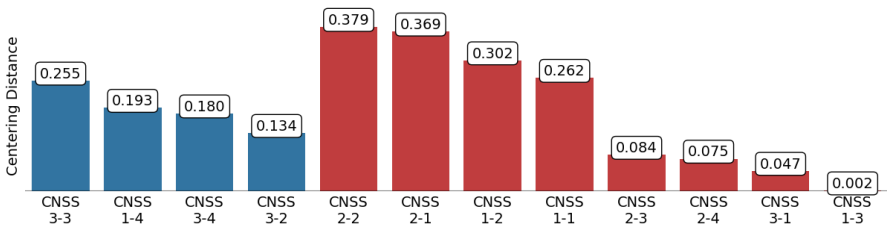


Figure 10: Centering distances of CNSS in descending order and grouped according to characterization.

After adjustments are made to the four CNSS that are operating predictably, the next step will be to investigate the remaining eight that are operating unpredictably. Unpredictable processes will continue to behave unpredictably until the assignable causes of exceptional variation that influence their behavior are eliminated. Subsequently, any efforts to center the eight CNSS that are operating unpredictably will waste resources and cause frustration. Centering is only valid when a process is operated predictably and processes only operate predictably when assignable causes are eliminated.

To get the other CNSS to operate predictably, the same critical thinking, investigatory, and analytical strategy that identified the excess lubricant issue

should be used. While the context of the effort will change in terms of the failure mode, the questions being asked and the analysis being performed remain the same. In due time, additional knowledge about the underlying causes of failures will be gained, changes will be made, and the iterative cycle of continual improvement will yield a CNSS manufacturing process that reliably produces products of world-class quality.

Summary

The challenge of manufacturing is the challenge of making multiples of parts and products that are functionally the same. This task is easier said than done. Regardless of the part or industry, manufacturing processes are always subject to variation. The CNSS lines and the needle-cannula subassemblies they produce are no different.

By using the network analysis framework as a guide, the manufacturing team responsible for the CNSS process was able to establish a holistic view. From this view, time and attention were directed to where they were needed most. Specifically, they were directed to evaluate and understand tip length failures. Extrapolated over the course of the year, tip length failures were estimated to cost the organization \$494,460 in scrap. By quantifying the occurrence of different tip length failure modes, the team was able to identify and address the dominant cost due to poor quality, excess lubricant, making this failure mode a thing of the past.

Although luck could have revealed this problem, luck does not constitute a repeatable method. Network analysis, however, is repeatable. The sequence of efforts guided by network analysis that eliminated the lubricant failure can now be replicated for other failure modes identified in Figure 2, like obstruction and leak test. Along the way, new knowledge will be gained about the mechanisms that generate scrap, and a more robust, cost-effective manufacturing process will emerge.